

Independent Reading Project

Endoscopic Vision and Surgical Scene Understanding for AI-Assisted Surgical Robotics

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Project Overview

This independent reading project reviews five representative studies on endoscopic visual perception, surgical instrument understanding, multimodal surgical AI, large vision models, and robot-assisted minimally invasive surgery. The selected papers cover deep learning-based endoscopic localization and 3D reconstruction, surgical instrument segmentation and tracking, grounded multimodal large language models for endoscopic surgery, large vision models for robot-assisted surgery, and multi-task benchmarking of surgical phase recognition, instrument keypoint estimation, and instrument instance segmentation.

The goal of this reading project is to build a structured understanding of how endoscopic vision and surgical scene understanding support AI-assisted surgical robotics. In particular, I focus on how visual perception, 3D reconstruction, surgical tool understanding, temporal modeling, and multimodal reasoning can contribute to robot-assisted navigation, visual servoing, surgical decision support, and human-robot collaboration in minimally invasive procedures.

Paper 1. Deep Learning-Based Localization, Mapping, and 3D Reconstruction for Endoscopy

Title: A Review of Deep Learning-Based Localization, Mapping and 3D Reconstruction for Endoscopy

Authors: Jiawei Zhong, Hongliang Ren, Qin Chen, Hui Zhang

Venue / Year: Journal of Micro and Bio Robotics, 2025

Topic: Deep learning-based localization, mapping, and 3D reconstruction for endoscopic navigation and scene understanding

Role in This Reading Project: Provides the foundation for understanding endoscopic localization, mapping, depth estimation, and 3D reconstruction.

Problem

This paper reviews the transition from traditional geometric endoscopic SLAM/SfM methods to deep learning-based frameworks for localization, mapping, and 3D reconstruction in challenging in-vivo environments. Traditional methods often struggle in endoscopic scenes because of low-texture tissues, illumination variation, specular reflection, tissue deformation, camera motion, and limited loop-closure opportunities.

Input / Output

- **Input:** Monocular or stereo endoscopic images/video sequences, and in some cases multimodal signals such as magnetic sensor data.
- **Output:** 6-DoF camera poses, dense depth maps, 3D point clouds, or reconstructed tissue surfaces.
- **Task:** Accurate localization, mapping, and 3D reconstruction for intra-operative guidance, endoscopic navigation, and robotic perception.

Method

The paper provides a taxonomy of deep learning-based endoscopic perception methods, including odometry estimation, 3D reconstruction, and joint estimation of camera pose and scene geometry. Representative modules include CNN-based visual encoders, recurrent or temporal modules for sequential modeling, GAN-based domain adaptation, Vision Transformers for global context representation, attention mechanisms for informative tissue or structural regions, and differentiable rendering or self-supervised reconstruction losses for geometry-aware learning.

The reviewed methods use public datasets such as EndoSLAM, SCARED, Hamlyn, UCL2019, SimCol, SERV-CT, and EndoMapper, as well as synthetic datasets and simulation environments, to mitigate the shortage of annotated clinical data. Common evaluation metrics include ATE and RPE for localization, AbsRel, SSIM, and RMSE for reconstruction or depth estimation, and FPS or inference runtime for real-time deployment.

Key Challenges

- **Annotated data scarcity:** Real clinical annotations are difficult to obtain, and there remains a domain gap between synthetic and real endoscopic data.
- **Dynamic and non-rigid scenes:** Tissue deformation, organ motion, tool occlusion, blood, bubbles, and specular reflection make robust perception difficult.
- **Illumination instability:** Endoscopic images often suffer from strong lighting variation and reflection.
- **Real-time constraints:** Surgical robotic systems require low-latency perception, while many deep learning models are computationally expensive.
- **Hardware limitations:** Some endoscopic or capsule-based systems have strict constraints on computation, size, and power consumption.

Relevance to AI-Assisted Surgical Robotics

Localization, depth estimation, and 3D reconstruction provide the perceptual foundation for AI-assisted surgical robotics. Accurate endoscopic perception can support robot-assisted navigation, surgical scene understanding, tool localization, visual servoing, intra-operative guidance, and safer manipulation in minimally invasive procedures.

Connection to My Background

This paper helps me connect my previous experience in embedded robotics and state estimation with endoscopic localization and reconstruction problems. My work with Kalman filtering, Bayes filtering, IMU/ToF sensing, and PID control provides useful background for understanding real-time perception-control challenges. In addition, my previous review work on endoscopic 3D reconstruction gives me an initial foundation for studying depth estimation, mapping, and 3D visual understanding in surgical scenes.

Possible Future Directions

- Explore pruning, quantization, and efficient architectures for real-time endoscopic depth estimation and reconstruction in resource-limited medical robotic systems.
- Study parameter-efficient fine-tuning methods such as LoRA-based adaptation to transfer general-purpose vision models to endoscopic depth estimation or scene understanding.
- Combine visual SLAM with IMU, magnetic sensing, or other sensor signals to improve robustness under illumination variation, deformation, and occlusion.
- Investigate how endoscopic depth estimation and localization can be integrated into robotic navigation, visual servoing, and safer surgical assistance.

Paper 2. Surgical Instrument Segmentation in Endoscopic Images and Videos

Title: Methods and Datasets for Segmentation of Minimally Invasive Surgical Instruments in Endoscopic Images and Videos: A Review of the State of the Art

Authors: Tobias Rueckert, Daniel Rueckert, Christoph Palm

Venue / Year: Computers in Biology and Medicine, 2024

Topic: Deep learning-based surgical instrument segmentation, tracking, and dataset analysis

Role in This Reading Project: Builds the technical basis for surgical instrument segmentation, tracking, and tool-aware robotic perception.

Problem

This paper reviews the development of surgical instrument segmentation from 2015 to 2023. It addresses the challenge of accurately identifying and localizing instruments in minimally invasive surgery, which is essential for downstream tasks such as robotic automation, surgeon skill assessment, safety monitoring, operating workflow optimization, and context-aware surgical assistance.

Input / Output

- **Input:** Endoscopic images or video sequences, often RGB frames from laparoscopic or robotic surgical systems.
- **Output:** Pixel-wise segmentation masks, including binary masks, category-wise masks, or part-level masks.

- **Task:** Transform raw endoscopic visual signals into structured semantic maps that distinguish surgical instruments from biological tissues and background structures.

Method

The paper systematically analyzes datasets and methods for semantic and instance-based segmentation of minimally invasive surgical instruments. The review covers both single-frame segmentation and approaches incorporating temporal information from successive video frames.

Representative methods include encoder-decoder architectures such as U-Net and DeepLab, attention mechanisms for focusing on small tool tips or thin structures, temporal modules such as LSTM or optical-flow-based methods for video consistency, and domain adaptation or synthetic-data strategies for improving generalization. The paper also discusses dataset development, including EndoVis 2015, EndoVis 2017, EndoVis 2018, EndoVis 2019, Kvasir Instrument, UCL dVRK, RoboTool, CholecSeg8k, HeiSurF, and AutoLaparo.

Common evaluation metrics include Dice coefficient, mean Intersection over Union (mIoU), and inference speed, especially when real-time deployment is required for surgical robotic systems.

Key Challenges

- **Domain gap:** Models trained on one dataset or simulation may not generalize to real clinical environments with different lighting, textures, tissue appearance, and instrument types.
- **Annotation cost:** Pixel-level medical video annotations require expert effort and are expensive to obtain.
- **Environmental noise:** Surgical scenes include smoke, blood, reflection, lens fogging, motion blur, and tool occlusion.
- **Instance-level difficulty:** Distinguishing multiple overlapping tools or visually similar instruments remains challenging.
- **Real-time requirements:** Segmentation models must often run at high frame rates to support closed-loop robotic assistance.

Relevance to AI-Assisted Surgical Robotics

Surgical instrument segmentation is a fundamental perception layer for robot-assisted surgery. Without precise segmentation, a robotic system cannot reliably track tools, avoid collisions, support visual servoing, provide augmented-reality overlays, or monitor surgical safety. It connects low-level visual perception to higher-level robotic assistance.

Connection to My Background

This paper connects naturally to my experience in computer vision and embedded robotic perception. My YOLOv11-based object detection project provides relevant preparation for understanding surgical tool localization and visual recognition, while this paper shows how detection-level perception can be extended to pixel-level and instance-level tool understanding. My background in endoscopic 3D reconstruction also suggests a possible connection between tool segmentation and geometry-aware scene reconstruction.

Possible Future Directions

- Adapt large vision or foundation models such as SAM-like architectures to surgical instrument segmentation with limited annotated data.
- Integrate robot kinematics with visual segmentation to handle tool occlusion and improve temporal consistency.
- Develop lightweight segmentation backbones for real-time closed-loop robotic assistance.
- Combine instrument segmentation with endoscopic depth estimation and 3D reconstruction for more robust surgical scene understanding.

Paper 3. Grounded Multimodal Large Language Model for Endoscopic Surgery

Title: EndoChat: Grounded Multimodal Large Language Model for Endoscopic Surgery

Authors: Guankun Wang, Long Bai, Junyi Wang, Kun Yuan, Zhen Li, Tianxu Jiang, Xiting He, Jinlin Wu, Zhen Chen, Zhen Lei, Hongbin Liu, Jiazheng Wang, Fan Zhang, Nicolas Padoy, Nassir Navab, Hongliang Ren

Venue / Year: Medical Image Analysis, 2026; available online in 2025

Topic: Surgical multimodal large language models, endoscopic scene understanding, visual grounding, and surgical dialogue

Role in This Reading Project: Introduces grounded multimodal surgical AI for endoscopic scene understanding and human-AI interaction.

Problem

This paper studies the limitations of general multimodal large language models in specialized endoscopic surgical environments. General MLLMs often suffer from hallucination, limited surgical-domain knowledge, and weak visual grounding ability, making them unreliable for identifying surgical instruments, tissues, anatomical structures, and local regions in endoscopic scenes. The paper aims to build a grounded surgical MLLM that can understand, localize, and converse about endoscopic surgical scenes more accurately.

Input / Output

- **Input:** Endoscopic images and natural-language instructions or questions.
- **Output:** Natural-language responses for surgical scene understanding and bounding-box coordinates for grounded visual localization.
- **Task:** Surgical scene understanding, visual question answering, region-based question answering, detailed image description, and visual grounding in endoscopic surgery.

Method

The paper proposes EndoChat, a grounded MLLM tailored for endoscopic surgery. It constructs a large-scale surgical instruction dataset, Surg-396K, and aligns endoscopic visual representations with a large language model to support multiple dialogue paradigms and surgical scene understanding tasks.

Key components include Surg-396K, a multimodal surgical instruction dataset; the Mixed Visual Token Engine (MVTE), which combines complementary features from DINOv2 and OpenCLIP to capture both spatial/geometric and semantic information; multi-scale visual encoding for global context and local anatomical or instrument details; and visual contrast-based reasoning or inference to reduce hallucination by encouraging the model to rely more on visual evidence than language priors.

Key Challenges

- **Domain gap:** Endoscopic images contain reflections, smoke, blood, tissue deformation, tool occlusion, and visual patterns that differ greatly from natural images.
- **Hallucination control:** The model must avoid generating plausible but visually unsupported surgical descriptions.
- **Grounding accuracy:** Surgical AI systems need to associate language responses with precise image regions.
- **Instruction data scarcity:** High-quality surgical dialogue and grounding annotations are difficult to obtain at scale.
- **Clinical reliability:** Surgical assistance requires high reliability, interpretability, and safety awareness.

Relevance to AI-Assisted Surgical Robotics

EndoChat provides a multimodal perception and interaction module for AI-assisted surgical robotic systems. For surgical robots to assist surgeons effectively, they need to understand surgical scenes, identify instruments and anatomical structures, localize important regions, and communicate information through natural language or structured outputs. This work points toward a possible interface between endoscopic visual perception, surgical decision support, and human-robot interaction in robot-assisted surgery.

Connection to My Background

This paper helps me understand how computer vision can be extended from object detection and reconstruction toward multimodal surgical scene understanding. My YOLOv11-based object detection experience is relevant to the visual recognition and localization aspects of this work, while my previous exposure to endoscopic 3D reconstruction helps me understand the importance of spatial perception in surgical scenes. Although EndoChat focuses on multimodal language-vision modeling, it motivates me to further study how surgical perception models can interact with robotic systems and human operators.

Possible Future Directions

- Extend grounded MLLMs from static images to surgical videos to understand procedural phases, tool-tissue interactions, and long-term surgical context.
- Integrate EndoChat-like models with endoscopic depth estimation or 3D reconstruction so that language instructions can be grounded in 3D space rather than only 2D bounding boxes.
- Explore quantization, distillation, or lightweight visual encoders to make surgical MLLMs more suitable for low-latency robotic assistance.
- Study how grounded surgical scene understanding can support robotic navigation, visual servoing, warning systems, and shared autonomy.

Paper 4. Large Vision Models for Robot-Assisted Surgery

Title: Innovating Robot-Assisted Surgery Through Large Vision Models

Authors: Zhe Min, Jiewen Lai, Hongliang Ren

Venue / Year: Nature Reviews Electrical Engineering, 2025

Topic: Large vision models, surgical foundation models, robot-assisted surgery, surgical perception, and embodied surgical intelligence

Role in This Reading Project: Provides a high-level roadmap for applying large vision models to upstream and downstream tasks in robot-assisted surgery.

Problem

This paper studies how large vision models can support the next stage of robot-assisted surgery by improving perception, scene understanding, simulation, planning, and decision support. It addresses the gap between general-purpose vision models and the specialized requirements of surgical environments, where models must handle domain-specific visual patterns, deformable anatomy, high safety requirements, limited annotated data, and real-time constraints.

Input / Output

- **Input:** Multimodal surgical data, including preoperative CT/MRI, intraoperative endoscopic images or videos, surgical instruments, robot kinematics, and potentially language or procedural information.
- **Output:** Surgical intelligence capabilities such as segmentation, detection, registration, surgical phase recognition, scene understanding, simulation, planning support, and robot-control-related decision outputs.
- **Task:** Connect low-level surgical perception tasks with higher-level surgical cognition, intraoperative decision support, simulation, and robot-assisted control.

Method / Main Framework

The paper presents a perspective on using large vision models and surgical foundation models to improve robot-assisted surgery. Instead of relying only on task-specific models, large-scale pretrained models can be adapted to surgical environments through transfer learning, self-supervised learning, multimodal learning, and generative modeling.

The main framework divides vision-related tasks in robot-assisted surgery into upstream and downstream tasks. Upstream tasks include classification, detection, segmentation, registration, depth estimation, and 3D reconstruction. Downstream tasks include surgical cognition, simulation, diagnosis, surgical navigation, decision support, and robot control. The paper argues that large vision models can contribute to both levels, but downstream surgical robotic intelligence requires further investigation because it involves multimodal data, physical constraints, and safety-critical decision making.

Key Challenges

- **Domain gap:** Natural-image foundation models do not directly transfer to surgical images because endoscopic scenes contain reflections, blood, smoke, occlusion, tissue deformation, and unusual textures.
- **Data scarcity:** High-quality labeled surgical data are difficult to collect because of privacy, annotation cost, clinical expertise requirements, and limited access.
- **Physical consistency:** Surgical environments involve deformable tissues and tool-tissue interaction, so perception and generative models need to respect physical and anatomical constraints.

- **Safety and reliability:** Surgical AI systems require uncertainty awareness, robustness, interpretability, and careful validation before clinical use.
- **Real-time deployment:** Robot-assisted surgery requires low-latency perception and decision support, while large models are often computationally expensive.

Relevance to AI-Assisted Surgical Robotics

This work provides a high-level roadmap for how large vision models may transform robot-assisted surgery from perception-only assistance toward more integrated surgical intelligence. It shows that AI-assisted surgical robots need visual recognition, scene understanding, multimodal reasoning, physical awareness, and eventually perception-control integration.

Connection to My Background

This paper helps me place my previous experience into a broader AI-assisted surgical robotics framework. My endoscopic 3D reconstruction review is related to upstream surgical perception tasks such as depth estimation, mapping, and registration. My YOLOv11-based object detection work is relevant to visual perception tasks such as instrument detection and localization. My embedded robotics experience with state estimation and control also helps me understand why perception outputs need to be reliable, real-time, and compatible with robotic navigation or assistance systems.

Possible Future Directions

- Explore model compression, quantization, distillation, or parameter-efficient fine-tuning to adapt large vision models for real-time endoscopic perception and surgical robotic systems.
- Combine large vision models with endoscopic depth estimation, SLAM, or 3D reconstruction so that surgical AI systems can reason about spatial structure rather than only 2D image regions.
- Study how anatomical constraints, tissue deformation, and tool-tissue interaction can be incorporated into perception or simulation models.
- Investigate how segmentation, detection, depth estimation, and surgical scene understanding can support robotic navigation, visual servoing, risk warning, and shared autonomy.

Paper 5. Surgical Phase Recognition, Instrument Keypoint Estimation, and Instance Segmentation in Endoscopy

Title: Comparative Validation of Surgical Phase Recognition, Instrument Keypoint Estimation, and Instrument Instance Segmentation in Endoscopy: Results of the PhaKIR 2024 Challenge

Authors: Tobias Rueckert, David Rauber, Raphaela Maerkl, Leonard Klausmann, Suemeyye R. Yildiran, Max Gutbrod, Danilo Weber Nunes, Alvaro Fernandez Moreno, Imanol Luengo, Danail Stoyanov, Nicolas Toussaint, Enki Cho, Hyeon Bae Kim, Oh Sung Choo, Ka Young Kim, Seong Tae Kim, Gonçalo Arantes, Kehan Song, Jianjun Zhu, Junchen Xiong, Tingyi Lin, Shunsuke Kikuchi, Hiroki Matsuzaki, Atsushi Kouno, João Renato Ribeiro Manesco, João Paulo Papa, Tae-Min Choi, Tae Kyeong Jeong, Juyoun Park, Oluwatosin Alabi, Meng Wei, Tom Vercauteren, Runzhi Wu, Mengya Xu, An Wang, Long Bai, Hongliang Ren, et al.

Venue / Year: Medical Image Analysis, 2026

Topic: Surgical phase recognition, instrument keypoint estimation, instrument instance segmentation, and surgical scene understanding

Role in This Reading Project: Connects surgical phase recognition, instrument keypoint estimation, and instance segmentation under a real-world multi-center benchmark.

Problem

This paper reports the results of the PhaKIR 2024 challenge, which targets reliable surgical scene understanding in endoscopic videos. It addresses three interrelated tasks in robot-assisted minimally invasive surgery: surgical phase recognition, instrument keypoint estimation, and instrument instance segmentation. These tasks are important because robust recognition and localization of surgical instruments are foundational for surgical training, skill assessment, autonomous assistance, and context-aware support in computer- and robot-assisted minimally invasive surgery.

Input / Output

- **Input:** Full-length laparoscopic cholecystectomy videos collected from multiple medical centers.
- **Output:** Surgical phase labels, pixel-wise instrument instance segmentation masks, and instrument keypoint coordinates.
- **Task:** Develop and benchmark temporally aware and context-driven algorithms for surgical phase recognition, instrument localization, and instrument segmentation in real-world endoscopic videos.

Dataset and Challenge Design

The PhaKIR challenge introduces a multi-center dataset containing thirteen full-length laparoscopic cholecystectomy videos collected from three medical institutions. The dataset provides unified annotations for three related tasks: surgical phase recognition, instrument keypoint estimation, and instrument instance segmentation. Unlike earlier datasets that often focus on a single task, single center, single frames, or simplified animal/synthetic data, PhaKIR supports joint investigation of surgical context and instrument localization using full real-world surgical sequences.

The paper follows the BIAS guidelines for biomedical image analysis challenges and provides a systematic comparison of submitted methods. The challenge encourages the use of temporal information across complete surgical procedures, rather than relying only on isolated frames.

Method / Evaluation Framework

The challenge evaluates submitted algorithms separately for the three tasks. Surgical phase recognition is evaluated using F1-score and balanced accuracy. Instrument instance segmentation is evaluated using Dice Similarity Coefficient, mAP based on IoU, and 95% Hausdorff Distance. Instrument keypoint estimation is evaluated using mAP based on Object Keypoint Similarity.

To improve evaluation robustness, the paper uses bootstrapping with 10,000 iterations and analyzes ranking stability across submitted methods. This provides not only raw performance comparison, but also insight into the reliability and consistency of methods under different sampled evaluations.

Key Findings

- **Temporal context is crucial for phase recognition:** All phase recognition submissions incorporated temporal information, showing its importance for surgical workflow understanding.
- **Transformer-based models are increasingly dominant:** Top-performing approaches in phase recognition and instance segmentation often used transformer-based architectures, such as specialized surgical transformers or Mask2Former-style models.
- **Instance segmentation remains challenging:** Even strong methods struggled to distinguish multiple instances of the same instrument class in cluttered surgical scenes.
- **Keypoint estimation is underexplored:** Only two teams participated in the keypoint estimation task, and overall performance remained limited.
- **Cross-center generalization remains difficult:** Methods performed better on videos from hospitals more represented in the training data, highlighting the challenge of domain shift across clinical environments.

Key Challenges

- **Real-world variability:** Endoscopic videos contain smoke, blur, bleeding, occlusion, rapid instrument motion, and varying visibility.
- **Domain shift across hospitals:** Different medical centers, imaging systems, surgical styles, and instrument distributions affect generalization.
- **Temporal modeling:** Temporal information is useful but remains underutilized in instrument segmentation and keypoint estimation.
- **Class imbalance:** Some instruments appear far less frequently than others, leading to poorer recognition and localization.
- **Instance-level ambiguity:** Overlapping or visually similar instruments make true instance segmentation and tracking difficult.

Relevance to AI-Assisted Surgical Robotics

This work is highly relevant to AI-assisted surgical robotics because it connects surgical scene understanding with practical robotic assistance needs. Surgical phase recognition provides procedural context; instrument segmentation enables

tool localization and scene parsing; and keypoint estimation supports pose understanding and distance estimation to risk structures. Together, these components can support context-aware warning systems, surgical navigation, autonomous endoscope guidance, skill assessment, and shared autonomy in robot-assisted surgery.

Connection to My Background

This paper helps me understand surgical scene understanding from a real-world benchmark perspective. My YOLOv11-based object detection experience is related to instrument localization and segmentation, while my embedded robotics experience with filtering, sensor fusion, and real-time control helps me appreciate the importance of temporal consistency and reliable perception. My previous exposure to endoscopic 3D reconstruction also connects to the spatial perception aspect of instrument localization and keypoint estimation.

Possible Future Directions

- Extend temporal modeling from surgical phase recognition to instrument instance segmentation and keypoint estimation.
- Jointly learn phase recognition, tool segmentation, keypoint estimation, and action understanding in a unified model.
- Develop domain adaptation or domain-invariant representation learning methods for more robust deployment across hospitals.
- Combine instrument segmentation and keypoint estimation with endoscopic depth estimation or 3D reconstruction.
- Design lightweight models that can support low-latency surgical navigation, visual servoing, and shared autonomy.

Overall Synthesis

Across the five papers, several common themes emerge. First, endoscopic vision is a core perception layer for AI-assisted surgical robotics, enabling localization, mapping, depth estimation, instrument segmentation, surgical workflow recognition, surgical scene understanding, and multimodal interaction. Second, surgical scenes differ substantially from natural images because of tissue deformation, specular reflection, occlusion, limited texture, class imbalance, limited annotations, and strict real-time constraints. Third, recent work is moving from isolated task-specific perception models toward integrated surgical intelligence, including large vision models, grounded multimodal models, multi-task surgical perception benchmarks, and perception-control integration.

This reading project also helps me identify how my previous experience can be connected to this research direction. My endoscopic 3D reconstruction review provides an entry point into surgical perception. My YOLOv11-based computer vision experience connects to instrument detection, segmentation, and localization. My embedded robotics work with Kalman filtering, Bayes filtering, PID control, sensor fusion, and real-time data logging connects to robotic perception and perception-control integration. Together, these experiences motivate me to further explore endoscopic vision, surgical scene understanding, and AI-assisted robotic perception for minimally invasive surgical systems.

Potential Research Directions

- **Endoscopic 3D perception and scene reconstruction:** Study depth estimation, SLAM, and 3D reconstruction methods that are robust to tissue deformation, reflection, and tool occlusion.
- **Surgical instrument segmentation and tool-aware perception:** Develop segmentation and tracking models that support tool localization, occlusion handling, and safer robotic assistance.
- **3D-aware surgical scene understanding:** Combine 2D visual grounding, segmentation, and multimodal reasoning with depth estimation or point-cloud reconstruction.
- **Temporal and multi-task surgical perception:** Jointly model surgical phase, instrument segmentation, keypoint localization, and tool-tissue interaction in real-world surgical videos.
- **Lightweight surgical AI models:** Explore compression, quantization, distillation, and efficient visual encoders for real-time endoscopic perception.
- **Perception-control integration:** Investigate how endoscopic perception outputs can support robotic navigation, visual servoing, risk warning, and shared autonomy.